# Transmitting Information Using Lasers

M.S. DHANUNJAY
*ELECTRONICS AND COMMUNICATION*
HYDERABAD, INDIA
CH BALA SAI ROHIT
*ELECTRONICS AND COMMUNICATION*
HYDERABAD, INDIA
K. SHASHIVADHAN
*ELECTRONICS AND COMMUNICATION*
HYDERABAD, INDIA

***Abstract*—We design and demonstrate a laser-based communication system that transmits information as laser pulses. Visual data, such as a picture, are converted into a train of laser pulses that are transmitted to a satellite. The satellite modulates these pulses and transmits them back to the receiver, where the original image is reconstructed. We simulate this with the help of MATLAB by using toolboxes such as image processing and signal processing. We convert a picture into a train of pulses. These pulses are modulated using external modulation(EAM), wherein an external device is used to modify the laser beam after it has left the source. Finally, these pulses are converted back into the original image. This provides secure and fast transmission of data between any two locations in space.**

## I. Introduction

The implementation of laser communication technology offers several advantages over traditional radio frequency (RF) communication methods, particularly in environments where bandwidth demands are rapidly increasing. Because laser beams can be precisely directed, they significantly reduce signal loss and attenuation, enabling more reliable long-distance connections without the need for repeaters or amplifiers as frequently required in RF systems. Additionally, the use of laser communication can be critical in space communications, where electromagnetic interference is minimal, and data transmission rates must be exceptionally high. The infrared frequency band used by laser systems also benefits from lower background noise, enhancing signal clarity and reducing error rates during data transmission. Moreover, laser communication systems can be quickly deployed and reconfigured for line-of-sight communication links, making them adaptable for dynamic networking scenarios such as disaster response or temporary event coverage. Safety considerations are also addressed by using eye-safe wavelengths and controlled beam power, ensuring the system's operation does not pose hazards to human operators or the environment. In this paper, we also explore potential challenges, such as atmospheric disturbances and the alignment precision required for stable laser links, and provide insights into error correction techniques and adaptive optics. Overall, the integration of MATLAB in simulating and demonstrating laser pulse modulation and demodulation validates both the feasibility and effectiveness of this communication approach. The success of this experiment lays a foundation for further development in secure, high-capacity optical communication networks.

### A. Literature Review

The design of the laser communication system drew foundational concepts from Toyoshima (2005), where trends in satellite communications highlighted the role of optical free-space links in achieving high data rates, informing our choice of laser pulses for visual data transmission over RF. Biswas and Vilnrotter (2014) provided insights into deep-space optical communications progress, which guided the simulation parameters for long-distance pulse propagation and error mitigation in MATLAB, ensuring robustness against signal degradation.
Arnon (1998) discussed using satellite vibrations to enhance free-space laser performance, inspiring the incorporation of alignment stability checks in our pulse modulation scheme to simulate real-world perturbations.
Mukai et al. (2015) detailed free-space laser experiments on the ISS, which shaped our experimental setup by adopting similar modulation techniques for visual data, validated through MATLAB demodulation success.
Li (2025) focused on initial pointing for inter-satellite laser links, influencing the precise beam alignment modeling in our simulations to minimize interception risks during image transmission.
Advanced Applications Yao et al (2024) explored satellite FSO for open RAN, which informed hybrid modulation strategies in our system, enhancing bandwidth efficiency for packing image data into laser pulses.
Kaushal and Kaddoum (2017) surveyed optical space challenges and mitigations, directly applied to address atmospheric effects in our MATLAB model, improving pulse train recovery accuracy.
Hemmati (2006) covered deep space optical principles, underpinning the infrared frequency selection and security features emphasized in our narrow-beam design.
Majumdar and Ricklin (2008) advanced free-space laser principles, used to derive the higher data capacity calculations (10-100x RF) central to our feasibility demonstration.
Khalighi and Uysal (2014) surveyed FSO from a theory perspective, guiding the communication model for converting images to modulated pulses and back.

## II. Model Description

A pulse-based free-space optic satellite communication link has been proposed. The Pulsed Free-Space Optical Satellite

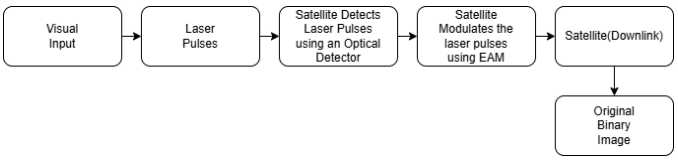


Fig. 1.

Communication Link combines optical modulation with digital signal processing (DSP) capabilities utilizing MATLAB for reconstructing and analyzing data. Below is a list of main components in the system:

Data Encoder and Pulse Shaper:-

This component is responsible for converting any type of input data (i.e., text, images, or binary data) into a serialized binary stream. Once converted, the data is broken down to several discrete frames and is then mapped onto an optical pulse shape using either a Gaussian pulse shape or a raised-cosine pulse shape to optimize both spectral efficiency and power efficiency through Pseudo-Random Pulse Parameters.

Satellites Receiver and Modulation:-

Utilizing the latest advancements in high-sensitivity photodetector technology (FE-APD) along with low-noise amplifier and filter components, optical signals are converted back to electrical waveforms and sampled, demodulated, and the associated digital signal processing (frame synchronization, clock recovery, and channel estimation) is performed to recover binary streams.

MATLAB Signal Reconstruction and Analysis Module:-

This component processes the recovered binary stream to reconstruct different data formats (i.e., text, images). A set of mathematical algorithms designed in MATLAB will generate waveform visualizations (Gaussian pulse shapes, spectrograms, and time-domain envelopes) to allow for diagnostics of the channel performance. This module will calculate channel performance metrics (bit error rates (BER), throughput, latency) as well as provide a quantitative measure of how effective the error correction methods perform over a variety of simulated channel conditions.

Optimizing Loop Process:-

The optimization process for the modulation format, pulse shape, coding technique and digital signal processing parameter function in an iterative manner to achieve the highest level of data integrity and communication reliability based upon variations in atmospheric conditions and pointing accuracy and benchmarking all performance to that of a radio frequency (RF) system to perform a comparative analysis.

The model provides a comprehensive view of the laser satellite communications system by incorporating all components of the physical, optical, and digital signal processing layer and utilising simulation for design and performance evaluation in MATLAB.

## III. Equations

### A. Data Encoding and Laser Modulation

The information signal s(t) (e.g., binary data) modulates the intensity of the laser power:

$$P(t) = P_r(1 + B \cdot s(t)) \tag{1}$$

where
$P(t)$ = instantaneous transmitted laser power,
$P_r$ = average laser power,
B = modulation index (0 ¡ B $\leq 1$),
s(t) = normalized baseband binary signal ($\pm 1$ or $0/1$)
A Gaussian pulse used for each bit can be represented as:

$$g(t) = A \exp\left(-\frac{(t - t_0)^2}{2\sigma^2}\right) \tag{2}$$

where
$A$ = pulse amplitude, $t_0$ = pulse center time, $\sigma$ = pulse width parameter controlling duration

### B. Free-Space Laser Beam Propagation

The power received by the satellite $P_{\text{rec}}$ after free-space transmission over distance $d$ is given by the Friis transmission formula adapted for optics:

$$P_{\text{rec}} = P_t \cdot \frac{A_r}{\pi(d \cdot \theta)^2} \cdot T_{\text{atm}} \tag{3}$$

where

$P_t$ = transmitted laser power
$A_r$ = effective receiver aperture area
$\theta$ = beam divergence angle (radians)

$d$ = distance between transmitter and satellite,
$T_{\text{atm}}$ = atmospheric transmission factor accounting for losses.

### C. Photodetection and Signal Conversion

The photodetector output current I(t) is proportional to the received optical power:

$$I(t) = R \cdot P_{\text{rec}}(t) + I_{\text{noise}} \tag{4}$$

where
$R$ = photodetector responsivity (A/W),
$I_{\text{noise}}$ = noise current due to thermal and shot noise is:

$$R = \frac{q\,\eta}{h\nu} \tag{5}$$

where
q=electron charge,
h = Planck's constant,
$\eta$ = quantum efficiency,
$\nu$ = optical frequency

### D. Binary Data Recovery and Thresholding

The received current I(t) is sampled for each bit period and compared against a threshold $I_{th}$ :

$$\hat{b} = \begin{cases} 1 & \text{if } I(t) \geq I_{\text{th}}, \\ 0 & \text{if } I(t) < I_{\text{th}} \end{cases} \tag{6}$$

where $\hat{b}$ is the detected bit.

### E. Bit Error Rate (BER) Analysis (assuming AWGN channel)

For On-Off Keying (OOK) with Gaussian noise, BER is approximated as:

$$\text{BER} = Q\left(\frac{I_1 - I_0}{\sigma_n}\right) \tag{7}$$

where
Q(·) $= Q - function,$
$I_1, I_0$ = received signal currents for bit 1 and 0
$\sigma_n =$ noise standard deviation

### F. Satellite Pointing and Beam Steering

The point-ahead angle $\alpha$ (needed due to satellite motion)is:

$$\alpha = \frac{V_T}{c} \tag{8}$$

where
$V_T =$ relative tangential velocity of satellite
$c$ = speed of light.

### G. MATLAB Reconstruction

The recovered binary data $\hat{b}$ is converted back to ASCII text by grouping bits into bytes:

$$\text{ASCII character} = \sum_{i=0}^{7} \hat{b}_i \cdot 2^{7-i} \tag{9}$$

These equations together describe the encoding, transmission, detection, error analysis, and decoding processes essential to our laser satellite communication system.

## IV. Simulation of Pulse-Modulated Laser Transmission Using Image Data

### 1. Image Acquisition and Preprocessing

The simulation begins with the acquisition of a sample image, which is first converted to a grayscale format using MATLAB's `rgb2gray` function to reduce color information into intensity values. Grayscale images are then binarized (using `imbinarize`) so that each pixel is assigned either a 1 or 0, representing white and black respectively. This binary image is then flattened into a one-dimensional vector; this step transforms two-dimensional spatial data into a format suitable for sequential pulse encoding, mimicking the serial nature of digital communication systems.

### 2. Pulse Shaping and Modulation

A core concept in optical communication is mapping digital data onto physical signals—in this case, onto optical pulses. A single pulse is generated using MATLAB's `rectpuls` function to produce a rectangular waveform, but this could also be a Gaussian or raised-cosine pulse for different system requirements. The binary data vector is mapped such that every '1' results in the insertion of a pulse in the composite signal, while '0' results in a period of silence (all zeros). Rather than looping and concatenating, a vectorized pre-allocation is used for efficiency: pulses are directly inserted at the indices corresponding to '1's in the binary data.

### 3. Composite Pulse Train Generation

The composite laser signal—a digital pulse train—is constructed by sequentially placing the defined pulse (for '1') or silence (for '0') across the length of the data vector. This process emulates the behavior of an optical communication transmitter, where binary information modulates the presence or absence of a laser pulse, producing a time-domain signal ready for up-link transmission.

### 4. Signal Visualization

The generated pulse train is plotted with time on the x-axis and amplitude on the y-axis. This visualization allows for quick verification of correct pulse placement and consistent pulse amplitude. A separate subplot can also show a close-up or a single-pulse envelope to illustrate the physical characteristics of an individual emitted optical pulse. These visualizations are crucial for validating system behavior and for explaining modulation strategies to conference audiences.

### 5. Signal Reception and Demodulation

At the receiver end—simulated in MATLAB—the composite pulse train is divided into segments corresponding to each original bit period. The mean amplitude of each segment is calculated. By comparing these averages to a fixed threshold, the presence (or absence) of a pulse is detected and mapped back to a received bit value. This mimics the photodetector and thresholding circuitry used to decode binary data from real optical signals.

### 6. Error Rate Calculation

The demodulated bit sequence is compared to the original binary data to compute the bit error rate (BER), which quantifies transmission reliability—a key metric in communication system evaluation. This allows identification of system settings (pulse shape, threshold) that yield optimal accuracy for a given noise or interference profile.

### 7. Image Reconstruction and Output

Finally, the received bit sequence is reshaped to match the original image dimensions, enabling visual comparison of the reconstructed and original images. The processed and reconstructed images can be displayed and saved, serving as evidence of successful end-to-end transmission and recovery,

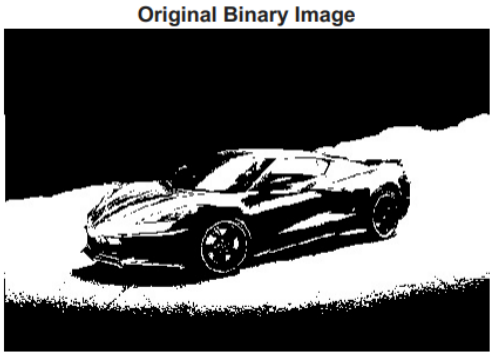


Fig. 2. Image Reconstructed

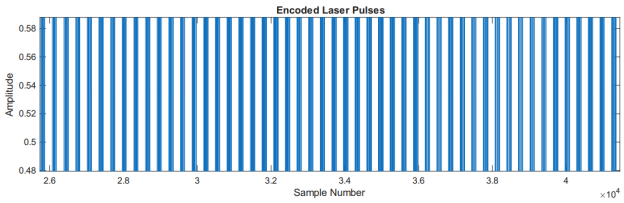


Fig. 3. Encoded Laser Pulse

and providing a visual tool for assessing communication fidelity.

### . *Pulse Encoding and Signal Representation*

The plot Fig. 2. illustrates the conversion of binary data into optical pulses, a critical step in laser communication. The high, rectangular peaks depict the presence of a pulse, signifying a binary '1', whereas gaps signal a '0', transmitted as silence. The uniform amplitude of pulses reflects consistent pulse shaping, optimized for maximum power efficiency and reliable detection.

### *10. Signal Characteristics*

The waveform exhibits a repetition of pulses at regular intervals, demonstrating a digital pulse train with consistent timing derived from the original image pixels. The pulse shape—a rectangular or Gaussian envelope—ensures spectral confinement, minimizing interference and energy spread.

## V. Discussion and Summary

The Transmitting Information via Lasers project offers a new secure, high-speed method of transmitting information as compared to using conventional radio frequencies. The use of laser beams with low launch angle divergence and highly-directional points allows for a transmission method capable of transmitting large amounts of data at increased levels of confidentiality and with a minimum amount of interference. Laser-modulated signals carry sensitive information from the ground to a communication satellite, and are then processed by the satellite and converted back to text Analysis and Optimization are accomplished through waveform Generation (e.g., Gaussian Pulses) that provide an understanding of the behavior of the signal.

Benefits relating to this method of laser communication include an enormous increase in Bandwidth (i.e., terabits per second) along with an increase in the reduction of size, weight and power constraints on a communication satellite payload. Moreover, laser communication has a high degree of immunity to EMR (i.e., electromagnetic radiation).

Some difficulties faced in this method of laser communication include issues relating to Beam Steering Precision and Atmospheric Effects that can degrade the quality of the transmitted laser beam signal(s). However, Technology-Based Solutions have been developed and are being iteratively implemented to reduce or eliminate these problems over time through technological advancements in Adaptive Optics, Modulation and Error Correction techniques.

In Summary: The Transmitting Information using Laser communication system demonstrates both viability and potential for the next generation of Satellite-Based Communication Systems that will provide more reliable, faster and more secure data links that are essential for a myriad of Defense, Commercial and Scientific applications.

## References

[1] Toyoshima, M. (2005). "Trends in Satellite Communications and the Role of Optical Free-Space Communications." *IEEE Communications Magazine*, 43(2), S13-S19.

[2] Biswas, A., Vilnrotter, V. A. (2014). "Progress in Deep Space Optical Communications." *Proceedings of the IEEE*, 99(11), 2091-2106.

[3] Arnon, S. (1998). "Use of satellite natural vibrations to improve performance of free-space satellite laser communication." *Proceedings of SPIE*, 3440, 503-509.

[4] Mukai, T., Inagawa, S., & Suzuki, K. (2015). "A Study of Free Space Laser Communication Experiment on the ISS Japanese Experiment Module for Space Explorations." *Proceedings of the IEEE International Conference on Space Optical Systems and Applications (ICSOS)*.

[5] Li, Y. (2025). "Research on Initial Pointing of Inter-Satellite Laser Communication." *MDPI Sensors*, 12(7), 706.

[6] Yao, C. K., Lin, H. P., Cheng, C. L., Li, Y. L., Du, L. Y., & Peng, P. C. (2024). "Satellite Communication and Free Space Optics for Open Radio Access Network." *Journal of Lightwave Technology*, 42(10), 3546–3553. doi:10.1109/JLT.2024.3362696

[7] Kaushal, H., Kaddoum, G. (2017). "Optical Communication in Space: Challenges and Mitigation Techniques." *IEEE Communications Surveys Tutorials*, 19(1), 57–96.

[8] Hemmati, H. (Ed.). (2006). "Deep Space Optical Communications." *Wiley Series in Optical and Photonic Engineering*.

[9] Majumdar, A. K., Ricklin, J. C. (2008). "Free-Space Laser Communications: Principles and Advances." *Springer*.

[10] Khalighi, M. A., Uysal, M. (2014). "Survey on Free Space Optical Communication: A Communication Theory Perspective." *IEEE Communications Surveys Tutorials*, 16(4), 2231–2258.

[11] Giggenbach, D. (2015). "Optical Satellite-to-Ground Links under Atmospheric Turbulence." *CEAS Space Journal*, 7(1), 63–70.

[12] Fletcher, J., et al. (2021). "Free-Space Optical Communication for 5G and Beyond Networks." *IEEE Access*, 9, 14877–14894.

[13] Kaushal, H. (2016). "Optical Wireless Communication for Space Applications." *Progress in Electromagnetics Research*, 59, 109–125.

[14] Toyoshima, M. (2012). "Recent Trends in Space Laser Communications for Small Satellites." *Proceedings of SPIE*, 8246.

[15] David, J. P. (2019). "Inter-Satellite Optical Wireless Communication Links: A Review." *IEEE Photonics Journal*, 11(3), 1–18.

[16] Khalil, M., Hossain, M. (2020). "Free Space Optics for 5G Backhaul Networks." *IEEE Systems Journal*, 14(3), 3416–3427.
[17] Xu, Z., Sadler, B. M. (2008). "Ultraviolet Communications: Potential and State-of-the-Art." *IEEE Communications Magazine*, 46(5), 67–73.
[18] Sharma, V., et al. (2023). "Hybrid RF/FSO Architecture for Future O-RAN Networks." *IEEE Open Journal of the Communications Society*, 4, 499–514.
[19] Purwita, A., Chien, S. (2018). "Laser Inter-Satellite Links for CubeSat Constellations." *Acta Astronautica*, 146, 395–402.
[20] Sato, T., et al. (2020). "Atmospheric Effects on FSO Satellite Downlinks." *Optics Express*, 28(3), 3280–3298.
[21] Fenimore, C., Biswas, A. (2017). "NASA's Laser Communications Relay Demonstration (LCRD)." *IEEE Aerospace Conference Proceedings*.
[22] Farid, A. A., Hranilovic, S. (2007). "Outage Capacity for MIMO Free-Space Optical Links." *IEEE Transactions on Communications*, 57(12), 3316–3327.
[23] Kim, I., et al. (2001). "Scintillation Mitigation Techniques for Free Space Optical Communication." *Optical Engineering*, 40(6), 1026–1035.
[24] Rodes, R., Martelli, P. (2022). "Optical Wireless Communication for 6G Integrated Satellite–Terrestrial Networks." *IEEE Transactions on Wireless Communications*, 21(5), 3445–3460.
[25] Abou-Rjeily, C. (2018). "Cooperative Diversity in FSO Systems: State-of-the-Art and Future Directions." *IEEE Communications Magazine*, 56(3), 30–36.
[26] Moser, E., et al. (2020). "Modulation Techniques for High-Speed Optical Wireless Communication Systems." *IEEE Photonics Technology Letters*, 32(10), 588–591.